

Creating habitats for beneficial species

Spatial scale: landscape

About this sustainable management strategy

Many beneficial organisms depend on small habitat features that provide food, nesting sites, shelter, or overwintering opportunities. In intensively managed vineyards, these resources are often limited, reducing the abundance of pollinators, natural enemies of pests, birds, reptiles, and other wildlife. Creating ecological niches within and around vineyards helps restore these resources and supports a wider range of species throughout the year.

Habitat features can be integrated into vineyard margins, unused corners, terraces, or other areas with limited production value. Flower-rich patches provide nectar and pollen for pollinators, while deadwood, stone piles, bare soil, and sand patches offer nesting and overwintering sites for insects, reptiles, and small mammals. Artificial structures such as bee hotels, bird nest boxes, bat boxes, or biodiversity towers can complement natural habitats where suitable nesting sites are scarce. Small ponds or seasonal wet areas further increase habitat diversity by supporting amphibians, insects, and birds. A combination of different habitat features distributed across the vineyard landscape generally supports a greater diversity of species than relying on a single intervention, while also strengthening ecological interactions that contribute to vineyard resilience.

Key Ecosystem Services:

- Habitat provision and biodiversity conservation
- Natural pest control (*support for beneficial insects and natural enemies*)
- Pollination
- Microclimate regulation (*reduced temperature extremes*)
- Cultural heritage (*preservation of traditional knowledge and historical land-use practices*)
- Landscape aesthetic (*distinct landscape character*)

Considerations for successful implementation

Placement within the vineyard landscape

Habitat features are most effective when located in areas of low production value, such as vineyard margins, terrace edges, field corners, or buffer zones, where they can complement existing semi-natural habitats without interfering with vineyard operations.

Regular maintenance and renewal

Some habitat features, particularly nesting boxes and insect hotels, require periodic inspection and maintenance to remain functional. Damaged materials, accumulated debris, or deteriorated nesting substrates should be replaced when necessary, while avoiding unnecessary disturbance during breeding or overwintering periods.

Use of natural materials

Natural, untreated materials such as wood, stone, reeds, and sand generally provide more suitable habitats and integrate better into the surrounding environment. Locally sourced materials also help maintain the ecological character of the landscape and reduce maintenance requirements.